

1. List 5 common BIOS or POST error messages you might see when starting a PC and write what hardware or configuration issue each one usually points to.

ANS:-

BIOS and POST errors appear when a computer detects a problem during startup. Some common error messages are:

No.	Error Message	What It Means
1	CMOS Battery Low	The CMOS battery is weak and needs to be replaced.
2	Keyboard Error	The keyboard is not connected properly or is not working.
3	Memory Test Failed	There may be a problem with the RAM or its connection.
4	Disk Boot Failure	The computer cannot find a bootable hard drive or operating system.
5	CPU Fan Error	The processor fan is not working properly or is not detected.

These messages help us identify hardware problems and fix them before the computer starts.

2. Given this scenario: On boot, a PC beeps 3 times in a repeating pattern and does not display anything on the screen. Research and write what this POST beep code typically means for an AMI BIOS, and suggest two troubleshooting steps.

ANS:-

In AMI BIOS, 3 short beeps usually indicate a base 64 KB memory failure. This means that the BIOS has detected a problem with the RAM (Random Access Memory).

The computer may not display anything because it cannot complete the memory test during startup.

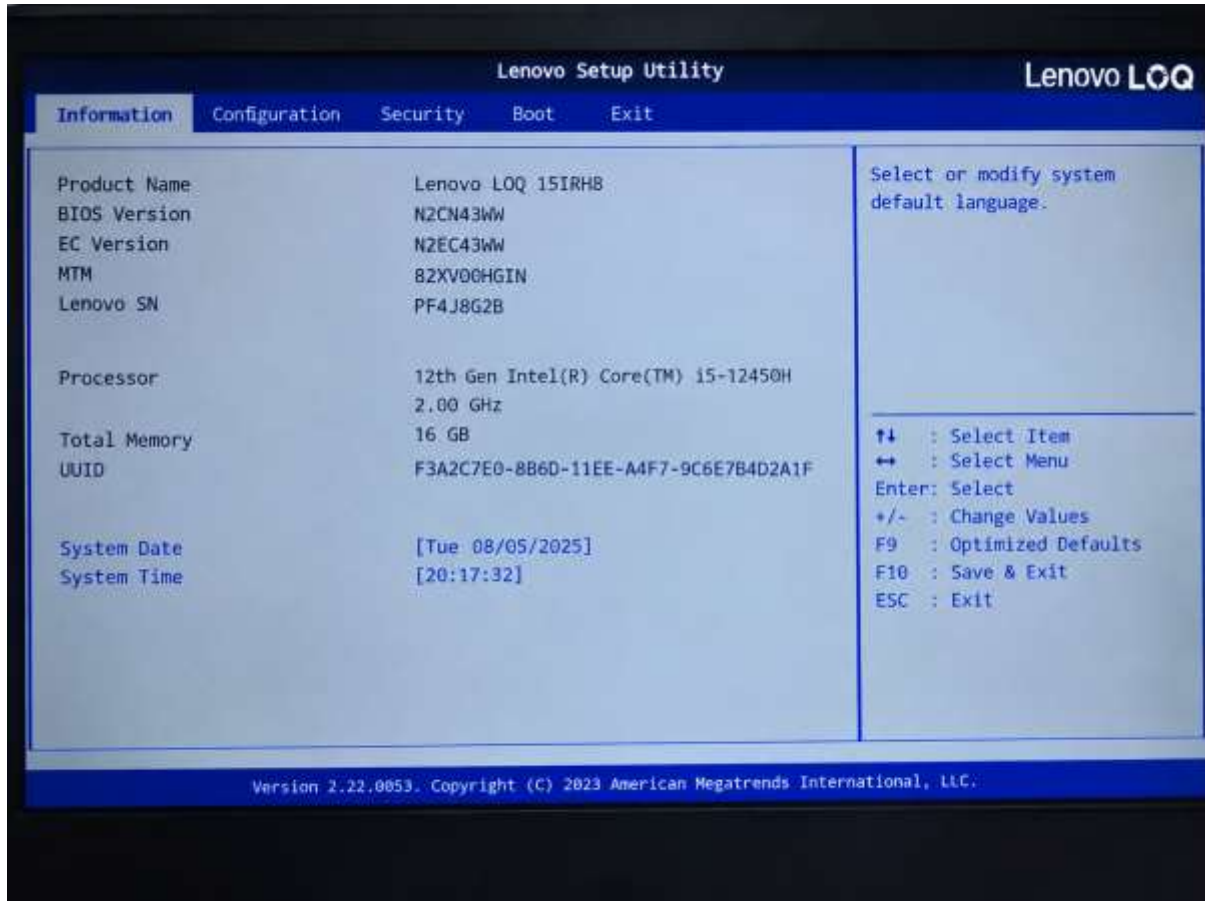
Two troubleshooting steps:

1. Check the RAM: Turn off the computer, remove the RAM stick, clean its contacts carefully, and insert it properly again.
2. Test the RAM: If the problem continues, try another RAM stick or test the existing RAM in a different slot to find the faulty part.

If the issue is still not solved, the motherboard may also need to be checked.

3. Access the BIOS setup utility on any available PC or virtual machine, take a screenshot of the main BIOS screen, and identify the motherboard manufacturer and BIOS version.

ANS:-



BIOS Information:

- Laptop Model: Lenovo LOQ 15IRH8
- Manufacturer: Lenovo
- BIOS Version: N2CN43WW
- EC Version: N2EC43WW
- Processor: 12th Gen Intel Core i5-12450H
- RAM: 16 GB
- BIOS Utility: Lenovo Setup Utility
- BIOS Firmware Provider: American Megatrends International, LLC.

4. You are given a screenshot of a POST error message that says 'CMOS checksum error – Defaults loaded.' Explain what this means and describe how you would resolve this issue in a real system.
Hint: Think about the CMOS battery and BIOS settings reset.

ANS:-

The message "CMOS checksum error – Defaults loaded" means that the BIOS has detected an error in the saved settings of the computer. Because of this, it has loaded the default settings automatically.

This problem usually occurs when the CMOS battery becomes weak or when the BIOS settings get corrupted or reset.

Steps to resolve the issue:

1. Turn off the computer and unplug the power cable.
2. Open the computer cabinet carefully.
3. Locate the CMOS battery on the motherboard. It is usually a small, round, coin-shaped battery.
4. Remove the old battery and replace it with a new one of the same type.
5. Start the computer and enter the BIOS setup.
6. Set the correct date and time and check the other BIOS settings.
7. Save the changes and restart the computer.

After replacing the battery and saving the settings, the error should normally disappear.

Conclusion: The CMOS battery helps the computer remember BIOS settings when it is switched off. Replacing a weak battery and restoring the correct settings usually solves this problem.